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ORIGINAL ARTICLE

Predominant role of the dopamine D_3 receptor subtype for mediating the quinpirole-induced inhibition of the vasopressor sympathetic outflow in pithed rats

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Abstract We have recently reported that quinpirole (a D_2-like receptor agonist) inhibits the vasopressor sympathetic outflow in pithed rats via sympatho-inhibitory D_2-like receptors. Since D_2-like receptors consist of D_2, D_3 and D_4 receptor subtypes, this study investigated whether these subtypes are involved in the above quinpirole-induced sympatho-inhibition by using antagonists of these receptor subtypes. One hundred fifty-six male Wistar rats were pithed and prepared for preganglionic spinal (T_7–T_9) stimulation of the vasopressor sympathetic outflow. This approach resulted in frequency-dependent vasopressor responses which were analysed before and during i.v. continuous infusions of either saline (0.02 ml/min) or quinpirole (1 μg/kg.min) in animals receiving i.v. bolus injections of vehicle [saline or dimethyl sulfoxide (DMSO)] or the antagonists L-741,626 (D_2), nafadotride or SB-277011-A (both D_3) as well as L-745,870 (D_4). Quinpirole inhibited the sympathetically-induced vasopressor responses. This sympatho-inhibition was (a) unaltered after 1 ml/kg saline, DMSO or 100 and 300 μg/kg L-741,626; (b) markedly blocked and abolished by, respectively, 30 and 100 μg/kg nafadotride or 100 and 300 μg/kg SB-277011-A and (c) slightly blocked after 30 and 100 μg/kg L-745,870, but 300 μg/kg L-745,870 produced no blockade whatsoever. Except for 300 μg/kg L-741,626 or 300 μg/kg L-745,870, the doses of the above compounds failed to modify per se the sympathetic vasopressor responses. The inhibition of the vasopressor sympathetic outflow induced by 1 μg/kg.min quinpirole in pithed rats is predominantly mediated by dopamine D_3 and, to a lesser extent, by D_4 receptor subtypes, with no evidence for the involvement of the D_2 subtype.

Keywords L-741,626 · L-745,870 · Nafadotride · SB-277011-A · Quinpirole · Vasopressor sympathetic outflow

Introduction

Dopamine is an important neuromediator that regulates numerous functions, including cardiovascular homeostasis. This molecule modulates cardiovascular function in the periphery through catecholamines release and hormone secretion (Missale et al. 1998). Furthermore, due to its regulatory action on the kidneys (Wang et al. 2008), adrenal glands (Damase-Michel et al. 1990) and vascular tone (Amenta et al. 2001), it contributes to blood pressure control. With the conjunction of structural, transductional and operational (pharmacological) criteria, dopamine receptors can be classified into D_1-like (which includes the D_1 and D_5 subtypes and activates G_s proteins) and D_2-like (which includes the D_2, D_3 and D_4 subtypes and activates $G_{i/o}$ proteins) types (Missale et al. 1998; Neve et al. 2004; Beaulieu and Gainetdinov 2011).

As far as the cardiovascular system is concerned, several lines of biochemical and pharmacological evidence have suggested that dopamine D_1-like receptors are mainly located on

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blood vessels (mediating direct vasodilatation), the heart (producing cardiac stimulation) and the kidney (inducing natriuresis on the proximal tubule), whereas D_2-like receptors are mainly found prejunctionally on perivascular and cardiac sympathetic nerves mediating sympatho-inhibition, which results in indirect vasodilatation and negative chronotropism, respectively (Willems et al. 1985; Missale et al. 1998; Amenta et al. 2000; Vaughan et al. 2000; Zeng et al. 2007).

Regarding peripheral sympatho-inhibition of the systemic vasculature, our group (Manrique-Maldonado et al. 2011) has recently shown the role of dopamine D_2-like receptors since (a) quinpirole (a D_2-like receptor agonist) dose-dependently inhibited the vasopressor responses produced by selective electrical stimulation of the preganglionic (T_7–T_9) sympathetic outflow (but not those to i.v. bolus injections of noradrenaline), whereas SKF-38393 (a D_1-like receptor agonist) failed to inhibit the sympathetically induced vasopressor responses, and (b) the vasopressor sympatho-inhibition by quinpirole remained unaltered after i.v. SCH-23390 (a D_1-like receptor antagonist) but was abolished after i.v. raclopride (a D_2-like receptor antagonist). On this basis, the present study in pithed rats has further investigated the specific role of the dopamine D_2, D_3 and D_4 receptor subtypes involved in the above sympatho-inhibition by quinpirole, analysing the effects of antagonists at D_2 (L-741,626), D_3 (nafadotride and SB-277011-A) as well as D_4 (L-745,870) receptor subtypes (see Table 1). Our results show that the dopamine D_2-like receptors inhibiting the vasopressor sympathetic outflow resemble the pharmacological profile of the D_3 receptor subtype and, to a lesser extent, of the D_4 receptor subtype, with no evidence for the involvement of the D_2 subtype.

Table 1 Binding affinity constants (pK_i) of quinpirole, L-741,626, nafadotride, SB-277011-A or L-745,870 for the different subtypes of dopamine receptors

Ligand	D_1-like		D_2-like		
	D_1	D_5	D_2	D_3	D_4
Agonist					
Quinpirole	5.7[a]	ND	8.3[a]	7.6[a]	7.5[a]
Antagonists					
L-741,626	ND	ND	**7.9**[b]	6.9[b]	6.1[b]
Nafadotride	ND	ND	7.9[c]	**9.3**[c]	6.5[c]
SB-277011-A			5.5[d]	**8.0**[d]	ND
L-745,870	ND	ND	5.7[e]	5.2[e]	**8.8**[e]

ND not determined

[a] Seeman and Van Tol (1994)

[b] Bowery et al. (1996)

[c] Audinot et al. (1998)

[d] Reavill et al. (2000)

[e] Patel et al. (1997)

Methods

Animals

All animal protocols in this investigation were approved by our Institutional Ethics Committee (CICUAL-Cinvestav), in accordance with the Guide for the Care and Use of Laboratory Animals in USA. Male Wistar normotensive rats (250–300 g) were maintained at a 12/12-h light/dark cycle (light beginning at 07:00 hours) and housed at constant temperature (22±2 °C) and humidity (50 %), with food and water ad libitum.

General methods

Experiments were carried out in a total of 156 rats. After quick anaesthesia with diethyl ether and cannulation of the trachea, the rats were pithed by inserting a stainless steel rod through the orbit and foramen magnum into the vertebral foramen (Shipley and Tilden 1947). Then, the animals were artificially ventilated with room air using a model 7025 Ugo Basile pump (56 strokes/min; stroke volume 20 ml/kg) (Kleinman and Radford 1964). After bilateral vagotomy, catheters were placed in (a) the left and right femoral veins, for the infusions of agonist and i.v. bolus injection of antagonists, respectively, and (b) the left carotid artery connected to a Grass pressure transducer (P23XL) for the recording of blood pressure. Heart rate was measured with a tachograph (7P4, Grass Instrument Co., Quincy, MA, USA), triggered from the blood pressure signal. Both parameters were recorded by a model 7 Grass polygraph (Grass Instrument Co., Quincy, MA, USA).

The pithing rod was replaced by an electrode enamelled except for 1 cm length, 9 cm from the tip, placing the uncovered segment at T_7–T_9 of the spinal cord to allow selective preganglionic stimulation of the vasopressor sympathetic outflow, as previously reported (Gillespie et al. 1970; Villalón et al. 1998). After 30 min, baseline values of diastolic blood pressure and heart rate were determined. Prior to electrical stimulation, the animals received gallamine (25 mg/kg, i.v.) to avoid electrically-induced muscular twitching. Ten minutes later, the animals were divided into two main sets (n=144 and 12, respectively). The first set of animals (n=144) was systematically pretreated with desipramine (50 μg/kg, i.v.) before each stimulus–response curve (S–R curve). Under these conditions, the vasopressor responses to lower stimulation frequencies are greater than those elicited without desipramine (Villalón et al. 1995a, b, 1998), and accordingly, the sympatho-inhibitory responses to quinpirole are particularly more pronounced at lower stimulation frequencies (Manrique-Maldonado et al. 2011). The second set of animals (n=12) did not receive desipramine and was performed for comparative purposes. After stable haemodynamic conditions had been maintained for at least 30 min, baseline values of diastolic blood pressure and heart rate were determined in both sets of animals.

The preganglionic vasopressor sympathetic outflow was then stimulated in both sets to elicit vasopressor responses by applying 10-s trains (2 ms monophasic rectangular pulses, 50 V), at increasing frequencies (0.03, 0.1, 0.3, 1 and 3 Hz) using a sequential schedule. When diastolic blood pressure had returned to baseline levels, the next frequency was applied. This procedure was performed systematically until the S–R curve had been completed (about 30 min). The body temperature of each pithed rat was maintained at 37 °C by a lamp and monitored with a rectal thermometer. At this point, the first set of animals was divided into three main groups (n=12, 66 and 66, respectively; 'Protocol 1'), while the second set was divided into two groups (n=6 each; 'Protocol 2').

Experimental protocols

Protocol 1

The first group (n=12) was subdivided into two subgroups (n=6 each) that received an i.v. continuous infusion of either physiological saline (two times, 0.02 ml/min) or quinpirole (1 μg/kg.min) by a WPI model sp100i pump (World Precision Instruments, Inc., Sarasota, FL, USA). Ten minutes after starting these infusions, a S–R curve was elicited as described above during the infusion of either compound to analyse their effects on the sympathetically induced vasopressor responses. It is important to note that these experiments were carried out in order to validate the reproducibility of the vasopressor responses and the maximal vasopressor sympatho-inhibition induced by 1 μg/kg.min quinpirole, as previously reported (Manrique-Maldonado et al. 2011).

The second group (n=66) was subdivided into 11 subgroups (n=6 each) that received an i.v. bolus injection of, respectively, (a) physiological saline (1 ml/kg); (b) DMSO 0.5 % (1 ml/kg); (c) 100 μg/kg L-741,626; (d) 300 μg/kg L-741,626; (e) 30 μg/kg nafadotride; (f) 100 μg/kg nafadotride; (g) 100 μg/kg SB-277011-A; (h) 300 μg/kg SB-277011-A; (i) 30 μg/kg L-745,870; (j) 100 μg/kg L-745,870 and (k) 100 μg/kg L-745,870. Twenty minutes thereafter, a S–R curve was elicited again, as described above, to analyse the effects of these compounds per se on the sympathetically-induced vasopressor responses.

The third group (n=66) was also subdivided into 11 subgroups (n=6 each) that received the same i.v. doses of the above compounds, as described for the second group. Ten minutes later, all subgroups received an i.v. continuous infusion of quinpirole (1 μg/kg.min). After 10 min, a S–R curve was again constructed during the infusion of quinpirole as described above, to analyse the effects of the aforementioned compounds on quinpirole-induced sympatho-inhibition.

Protocol 2

The two groups of this protocol (n=6 each) received an i.v. continuous infusion of, respectively, physiological saline (two times, 0.02 ml/min) and quinpirole (1 μg/kg.min) as previously described. Ten minutes after starting these infusions, a S–R curve was elicited as described above during the infusion of either compound to analyse their effects on the sympathetically-induced vasopressor responses without desipramine. It is important to note that these experiments were performed in order to ascertain that activation of sympatho-inhibitory receptors stimulated by quinpirole is operative also in the absence of desipramine, as previously reported for serotonin (Villalón et al. 1995a).

The doses of saline and quinpirole were infused at 0.02 ml/min for about 50 min, by which time the S–R curves had been completed. The doses of these compounds were selected on the basis of previous studies in which reproducible vasopressor responses after saline and consistent inhibitory effects by quinpirole on the S–R curves had been elicited with no changes in baseline diastolic blood pressure or heart rate (Manrique-Maldonado et al. 2011). The administration of all compounds was sequential. The interval between the different stimulation frequencies was dependent on the duration of the resulting vasopressor responses (3–15 min), as we waited until diastolic blood pressure had returned to baseline values.

Compounds

Apart from the anaesthetic (diethyl ether), the compounds used in the present study (all obtained from Sigma Chemical Co., St. Louis, MO, USA) were the following: gallamine triethiodide, desipramine hydrochloride, quinpirole dihydrochloride, (±)-3-[4-(4-chlorophenyl)-4-hydroxypiperidinyl] methylindole (L-741,626), nafadotride tartrate, *N*-[*trans*-4-[2-(6-cyano-3,4-dihydroisoquinolin-2(1*H*)-yl)ethyl]cyclohexyl]quinoline-4-carboxamide hydrochloride (SB-277011-A) and 3-[[4-(4-chlorophenyl)piperazin-1-yl]methyl]-1*H*-pyrrolo[2,3-*b*]pyridine hydrochloride (L-745,870). Desipramine, gallamine, quinpirole, nafadotride and L-745,870 were dissolved in physiological saline, whereas L-741,626 and SB-277011-A were dissolved in DMSO 0.5 % (v/v) in saline. These vehicles had no effect on baseline values of diastolic blood pressure and heart rate (not shown). Fresh solutions were prepared for each experiment. The doses of compounds mentioned in this text refer to their free base except in the case of desipramine and gallamine, where they refer to their respective salts.

Data presentation and statistical evaluation

All data in the text and figures are presented as mean ± SEM. The peak changes in diastolic blood pressure produced by electrical stimulation were expressed as percent change from baseline. The difference between the vasopressor responses within one subgroup of animals was evaluated with Student–Newman–Keuls post hoc test, once a two-way repeated measures analysis of variance (randomised block design) had revealed that the samples represented different populations (see Steel and Torrie 1980). Statistical significance was accepted at $p<0.05$ (two tailed).

Results

Systemic haemodynamic variables

The baseline values of diastolic blood pressure and heart rate in the rats with and without desipramine were, respectively, 55±5 mmHg and 324±7 beats/min ($n=144$) and 59±8 mmHg and 319±5 beats/min ($n=12$). After the first i.v. injection of desipramine (50 μg/kg), both variables increased but returned to baseline values after 5 min (i.e. 60±6 mmHg and 326±8 beats/min; $n=144$). These latter values in desipramine-pretreated rats were not significantly modified (data not shown) as follows: (a) during the i.v. continuous infusions of physiological saline (0.02 ml/min) or quinpirole (1 μg/kg.min); (b) after the i.v. bolus injections of saline (1 ml/kg), DMSO (1 ml/kg), L-741,626 (100 and 300 μg/kg), nafadotride (30 and 100 μg/kg), SB-277011-A (100 and 300 μg/kg) or L-745,870 (30, 100 and 300 μg/kg) or (c) the subsequent i.v. treatments with desipramine (50 μg/kg each) before each S–R curve.

Initial effects produced by electrical stimulation of the preganglionic (T_7–T_9) sympathetic nerves on blood pressure and heart rate

The onset of the responses induced by electrical stimulation of the vasopressor sympathetic outflow (0.03–3 Hz) in rats with and without desipramine was immediate and resulted in frequency-dependent increases in diastolic blood pressure (see control responses in Figs. 1, 2 and 3). It must be emphasised that these vasopressor responses (a) were significant ($p<0.05$) when compared with their corresponding baseline values and (b) involved selective sympathetic stimulation of the systemic vasculature since only negligible (if any) changes in heart rate were observed (not shown), as previously reported by our group (Manrique-Maldonado et al. 2011).

Effect of i.v. continuous infusions of physiological saline or quinpirole on the sympathetically-induced vasopressor responses in the presence or in the absence of desipramine

Figure 1 shows the vasopressor responses induced by stimulation of the vasopressor sympathetic outflow before (control) and during the i.v. continuous infusion of physiological saline (0.02 ml/min) or quinpirole (1 μg/kg.min) in rats with and without desipramine. In animals infused with vehicle (saline), the sympathetically-induced vasopressor responses remained essentially unchanged when repeating two subsequent S–R curves (Fig. 1a, c). In contrast, the infusion of quinpirole produced a significant inhibition of the sympathetically-induced vasopressor responses (Fig. 1b, d) at all stimulation frequencies (0.03–3 Hz), with this inhibition being less prominent in the rats without desipramine. The above results with desipramine reconfirm those previously reported by Manrique-Maldonado et al. (2011), in which 1 μg/kg.min quinpirole inhibited the vasopressor sympathetic outflow at all stimulation frequencies, with higher doses producing a supramaximal effect. That is why 1 μg/kg.min quinpirole was chosen for further pharmacological analysis in rats with desipramine (see below).

Effect of vehicles as well as several doses of L-741,626, nafadotride, SB-277011-A or L-745,870 per se on the sympathetically-induced vasopressor responses

Figure 2 shows that, as compared to the control S–R curve, the sympathetically-induced vasopressor responses remained unchanged after an i.v. bolus injection of (a) 1 ml/kg saline (Fig. 2a); (b) 1 ml/kg DMSO 0.5 % (Fig. 2b); (c) 100 μg/kg L-741,626 (Fig 2c); (d) 30 and 100 μg/kg nafadotride (Fig. 2e, f); (e) 100 and 300 μg/kg SB-277011-A (Fig. 2g, h) and (f) 30 and 100 μg/kg L-745,870 (Fig. 2i, j). However, 300 μg/kg L-741,626 (Fig. 2d) or 300 μg/kg L-745,870 (Fig. 2k) produced a significant attenuation of the vasopressor responses, except at the lowest stimulation frequencies.

Effect of vehicles as well as several doses of L-741,626, nafadotride, SB-277011-A or L-745,870 on the sympatho-inhibition produced by quinpirole

Figure 3 shows the effects produced after i.v. administration of the above compounds (at the same doses as in Fig. 2) on the inhibition of sympathetically-induced vasopressor responses induced by quinpirole. Consistent with previous findings (Manrique-Maldonado et al. 2011), the i.v. continuous infusion of quinpirole elicited a reproducible sympatho-inhibition in the animals treated with an i.v. bolus injection of 1 ml/kg saline (Fig. 3a). This quinpirole-induced sympatho-inhibition (a) remained also unchanged in the animals treated with i.v. bolus injections of 1 ml/kg DMSO 0.5 % (Fig. 3b) or

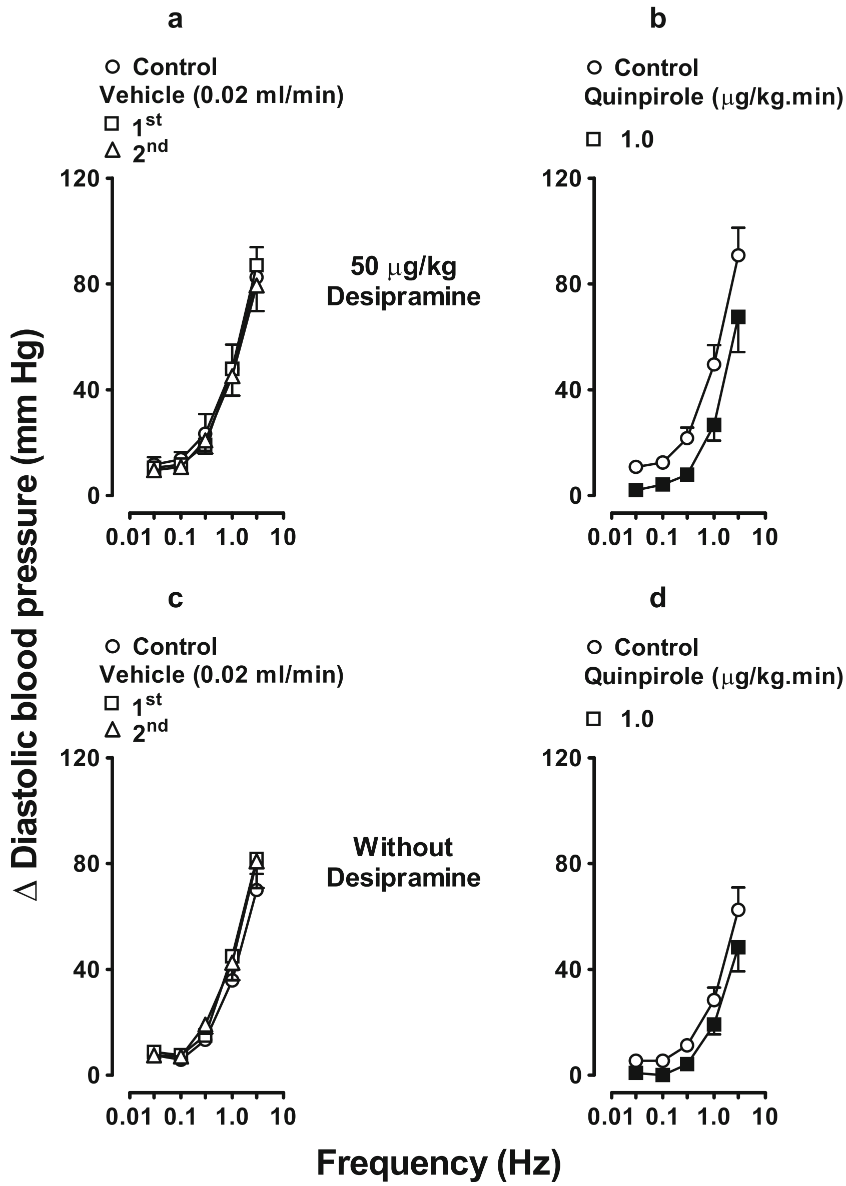

Fig. 1 Effects of an i.v. infusion of **a**, **c** vehicle (saline, 0.02 ml/min, two times, n=6) or **b**, **d** quinpirole (1 μg/kg.min) in animals in the presence (*upper panel*) or in the absence (*lower panel*) of desipramine (50 μg/kg, i.v.) on the electrically-induced vasopressor responses. *Empty symbols* depict control (*empty circle*) or non-significant (p> 0.05) responses (*empty square*, *empty triangle*). *Filled square* represents significantly different responses (p<0.05) vs. control. Δ Diastolic blood pressure stands for "increase in diastolic blood pressure"

100 and 300 μg/kg L-741,626 (Fig. 3c, d, respectively); (b) was markedly blocked and abolished by, respectively, 30 and 100 μg/kg nafadotride (Fig. 3e, f) or 100 and 300 μg/kg SB-277011-A (Fig. 3g, h) and (c) partly blocked by 30 and 100 μg/kg L-745,870 (particularly at the lowest stimulation frequencies; Fig. 3i, j) but, interestingly, 300 μg/kg L-745,870 produced no apparent blockade (Fig. 3k).

Discussion

General

Apart from the implications discussed below, our results show, in the first instance, that the inhibition of the vasopressor sympathetic outflow induced by 1 μg/kg.min quinpirole in pithed rats is operative even in the absence of desipramine (compare Fig. 1b, with d). Admittedly, the inhibition by quinpirole was less prominent in the latter condition but better approximates the physiological situation; thus, inhibition of neuronal noradrenaline reuptake mechanisms by desipramine in our experiments is not an obligatory prerequisite to show the inhibitory effect of quinpirole. Moreover, our study demonstrates that quinpirole-induced inhibition of the vasopressor sympathetic outflow is predominantly mediated by the dopamine D_3 (completely blocked by nafadotride or SB-277011-A) and, to a lesser extent (if any), by the D_4 (partly blocked by L-745,870) receptor subtypes, with no pharmacological evidence for the involvement of the D_2 subtype (resistant to blockade by L-741,626). From a transductional point of view, D_2-like (i.e. D_2, D_3 and D_4) receptor signalling is mediated by the

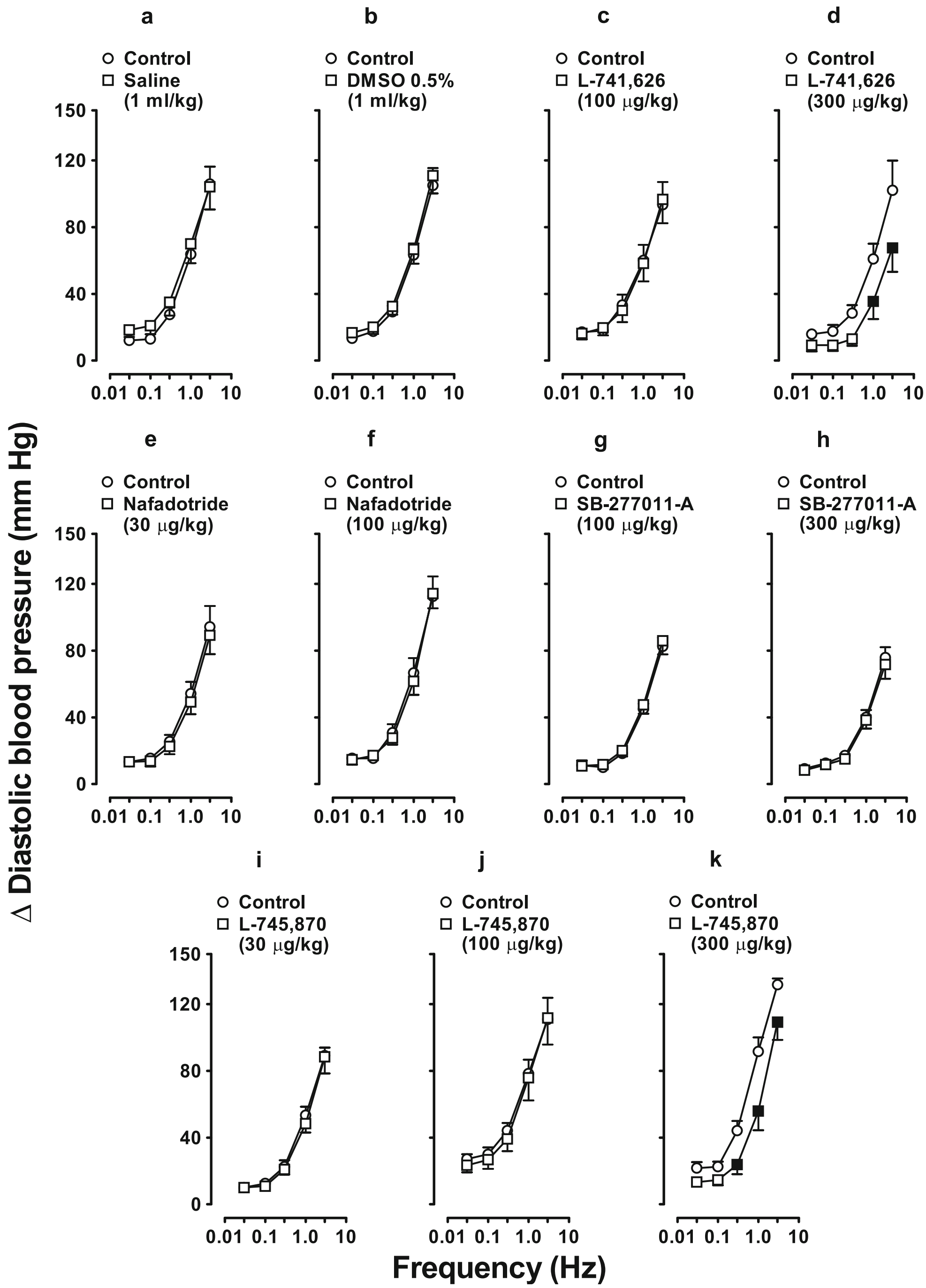

Fig. 2 Effect per se of i.v. bolus injections of saline (**a** 1 ml/kg), DMSO 0.5 % (**b** 1 ml/kg), L-741,626 (**c** 100 µg/kg, **d** 300 µg/kg), nafadotride (**e** 30 µg/kg, **f** 100 µg/kg), SB-277011-A (**g** 100 µg/kg, **h** 300 µg/kg) and L-745,870 (**i** 30 µg/kg, **j** 100 µg/kg, **k** 300 µg/kg) (n=6 each) on the sympathetically-induced vasopressor responses. *Empty symbols* depict control (*empty circle*) or non-significant (*empty square*) responses. *Filled square* represents significantly different responses (p<0.05) vs. control. Δ Diastolic blood pressure stands for "increase in diastolic blood pressure"

heterotrimeric $G_{i/o}$ proteins that, amongst other effects, inhibit adenylyl cyclase activity, inactivate Ca^{2+} channels and/or activate inwardly rectifying K^+ channels (Neve et al. 2004; Beaulieu and Gainetdinov 2011). These are signal transduction systems usually associated with sympatho-inhibition (Boehm and Kubista 2002; De Jong and Verhage 2009), although, admittedly, we have no direct evidence that any of these mechanisms is involved in our experimental model. Moreover, we did not measure sympathetic nerve activity directly, but the electrically-induced neurotransmitter release in our study was estimated indirectly by measurement of the evoked vasopressor response, as previously mentioned by our group (Villalón et al. 1995a, b, 1998). Hence, the responses to 1 µg/kg.min quinpirole were considered to be sympatho-inhibitory since this agonist inhibited the vasopressor responses induced by stimulation of the preganglionic (T_7–T_9) sympathetic outflow in rats with and without desipramine (Fig. 1), without affecting those by exogenous noradrenaline (Manrique-Maldonado et al. 2011).

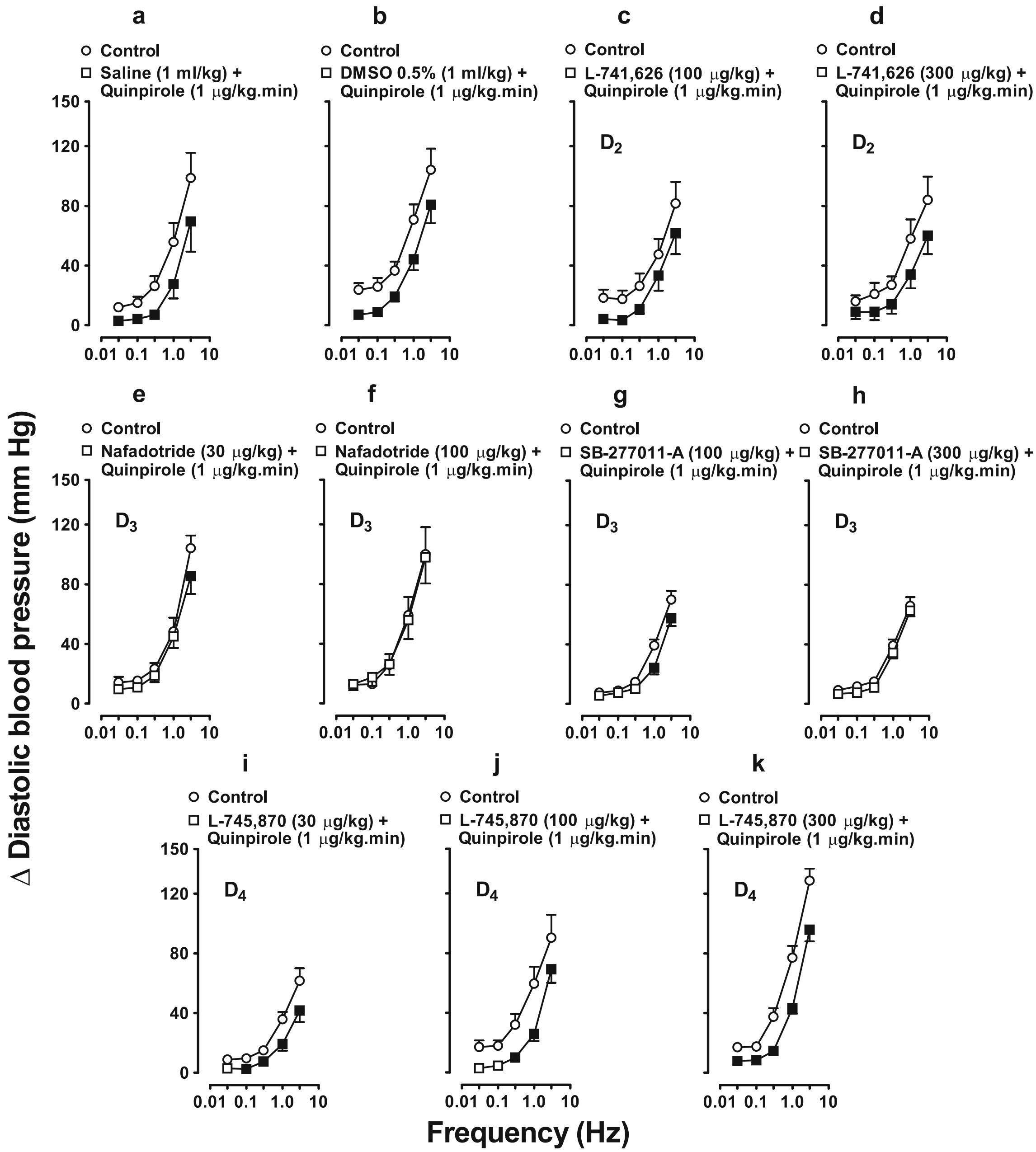

Fig. 3 Effects of i.v. bolus injections of saline (**a** 1 ml/kg), DMSO 0.5 % (**b** 1 ml/kg), L-741,626 (**c** 100 μg/kg, **d** 300 μg/kg), nafadotride (**e** 30 μg/kg, **f** 100 μg/kg), SB-277011-A (**g** 100 μg/kg, **h** 300 μg/kg) and L-745,870 (**i** 30 μg/kg, **j** 100 μg/kg, **k** 300 μg/kg) (n=6 each) on quinpirole-induced sympatho-inhibition (1 μg/kg.min, i.v.). Antagonists were administered 10 min before the quinpirole infusion started. *Empty symbols* depict control (*empty circle*) or non-significant (*empty square*) responses. *Filled square* represents significantly different responses (p<0.05) vs. control. Δ Diastolic blood pressure stands for "increase in diastolic blood pressure"

Systemic haemodynamic changes

The fact that diastolic blood pressure and heart rate were transiently increased (p<0.05) after desipramine can be attributed to an inhibition of noradrenaline reuptake (Bechtel et al. 1986). Moreover, the potentiation of the sympathetically-induced vasopressor responses after desipramine (see Villalón et al. 1995a; Fig. 1) revealed more pronounced sympatho-inhibitory responses to quinpirole (compare Fig. 1b, with d; Manrique-Maldonado et al. 2011). This sympatho-inhibition by quinpirole is not due to tachyphylaxis of the sympathetically induced vasopressor responses as the corresponding S–R curves remained unchanged after saline (Fig. 1a, c). An additional conclusion drawn from this finding is that no time-dependent changes occurred during the course of our experimental protocols.

On the other hand, it is noteworthy that the sympathetically-induced vasopressor responses remained unchanged after an i.v. bolus injection of the following: (a) 1 ml/kg saline; (b) 1 ml/kg DMSO 0.5 %; (c) 100 μg/kg L-741,626; (d) 30 and 100 μg/kg nafadotride; (e) 100 and 300 μg/kg SB-277011-A and (f) 30 and 100 μg/kg L-745,870 (see Fig. 2). These findings indicate that the above compounds, at the doses used, were essentially devoid of any effects per se on the sympathetically-induced vasopressor responses. Interestingly, the highest doses (300 μg/kg) of L-741,626 (Fig. 2d) and L-745,870 (Fig. 2k) produced a significant attenuation of the sympathetically-induced vasopressor responses, except at the lowest stimulation frequencies. Amongst other possible explanations, this attenuation may have been due to the following: (a) the moderate binding affinity of L-741,626 (pK_i 6.6; Millan et al. 2000) and L-745,870 (pIC_{50}<6.5; IC_{50}<300 nmol/l; Patel et al. 1997) for α_1-adrenoceptors; (b) the major role of vascular α_1-adrenoceptors in the rat sympathetic vasopressor responses (Flavahan et al. 1985; Bulloch and McGrath 1988) and (c) a more evident blockade when the vasopressor responses are greater in magnitude (i.e. the higher the stimulation frequencies the greater the vasopressor responses and, consequently, the more evident the blockade). On this basis, 300 μg/kg of either L-741,626 or L-745,870 may have produced a moderate blockade of vascular α_1-adrenoceptors in resistance blood vessels which, in turn, resulted in an attenuation of the sympathetic vasopressor responses (Docherty 2011).

Furthermore, since all of the above compounds failed to modify the baseline values of diastolic blood pressure and heart rate, any effect of a given antagonist on quinpirole-induced sympatho-inhibition (at doses that did not show any effect per se on the sympathetic vasopressor responses) is presumably attributable to a direct interaction of the antagonist with its respective receptors on the perivascular sympathetic nerves, rather than to changes in (i.e. attenuation of) the sympathetic vasopressor responses. Although 300 μg/kg of either L-741,626 or L-745,870 attenuated these vasopressor responses (see above), these doses were not excluded from our experimental protocols because the vasopressor responses induced by the lower stimulation frequencies remained unchanged (Fig. 2d, k), and this could provide additional information on the possible role of corresponding receptor subtypes.

Possible role of dopamine D_2, D_3 and D_4 receptor subtypes in quinpirole-induced inhibition of the sympathetic vasopressor responses

Our results showing that quinpirole-induced sympatho-inhibition remained unchanged after saline (compare Fig. 3a with Fig. 1b) or DMSO 0.5 % (compare Fig. 3b with Fig. 1b or with Fig. 3a) indicate that this response is reproducible and not affected by these vehicles (used to dissolve the antagonists). Considering the classification criteria for dopamine D_2-like receptors (which include the D_2, D_3 and D_4 subtypes) (Missale et al. 1998; Neve et al. 2004; Beaulieu and Gainetdinov 2011), the D_2 (L-741,626) and D_3 (nafadotride) preferring receptor subtype antagonists as well as the D_3-selective (SB-277011-A) and D_4-selective (L-745,870) receptor subtype antagonists were used in doses that, on the basis of their affinities (Table 1), are considered high enough to block their respective subtypes. Although the possible interference by some pharmacokinetic factors involved in the i.v. administration (i.e. metabolism and excretion) in the pithed rat cannot be categorically excluded, the fact that quinpirole-induced sympatho-inhibition was (see Fig. 3) (a) unchanged after 100 and 300 μg/kg L-741,626 does not seem to support the role of the D_2 receptor subtype; (b) markedly blocked and abolished by, respectively, 30 and 100 μg/kg nafadotride or 100 and 300 μg/kg SB-277011-A suggests the involvement of the D_3 receptor subtype and (c) partly blocked by 30 and 100 μg/kg L-745,870 suggests, to a lesser extent (if any, and particularly at the lowest stimulation frequencies), the role of the D_4 receptor subtype. It must be emphasised that, based on their binding affinities (Table 1), the doses used of the above antagonists were high enough to block (even completely at their highest doses) their respective receptors. However, the profile of blockade (or lack of blockade) displayed by these antagonists in our study deserves further considerations in view of their binding properties (Table 1) and effects per se on the sympathetic vasopressor responses (Fig. 2).

As far as L-741,626 is concerned, it may be argued that, probably, higher doses of this compound could have blocked the sympatho-inhibition by quinpirole. Nevertheless, L-741,626 also displays a reasonable affinity (pK_i 6.9) for the D_3 subtype and a moderate affinity (pK_i 6.1) for the D_4 subtype (Table 1), so that higher doses could have also blocked these receptors. In addition, 300 μg/kg of this compound attenuated per se the vasopressor responses at the two highest stimulation frequencies (Fig. 2d), and this may have overshadowed any blockade of the response to quinpirole (particularly at the highest stimulation frequencies; Fig. 3d). Notwithstanding, if the D_2 subtype were involved, 100 μg/kg L-741,626 should have produced, at least, a partial blockade of the response to quinpirole, but this was not the case (see Fig. 3c). Indeed, 100 and 300 μg/kg L-741,626 markedly blocked (unpublished observations) and abolished (Villalón et al. 2011), respectively, quinpirole-induced cardiac sympatho-inhibition in pithed rats, a response mediated by the D_2, but not the D_3 or D_4, receptor subtypes.

Moreover, nafadotride not only displays a very high affinity (pK_i 9.3) for the D_3 subtype but also a considerable affinity (pK_i 7.9) for the D_2 subtype (see Table 1). Hence, that the response to quinpirole was potently blocked by nafadotride (Fig. 3e, f) suggests, in its own right, the role of the D_2 and D_3 subtypes, although the D_2 subtype was excluded based on the inactivity of L-741,626 (see above). To make the role of the D_3 subtype even more convincing, we decided to use a selective D_3 receptor antagonist, namely, SB-277011-A (pK_i 5.5 and 8.0 for the D_2 and D_3 subtypes, respectively; Table 1). Since 100 and 300 μg/kg SB-277011-A markedly blocked and abolished the response to quinpirole (Fig. 3g, h), these findings suggest once again the role of the D_3 subtype, although, admittedly, the free plasma concentrations of these (and the other) antagonists were not determined in our experimental model.

Regarding L-745,870, it is noteworthy that 300 μg/kg L-745,870 produced no apparent blockade on the response to quinpirole (Fig. 3k), unlike its lower doses (Fig. 3i, j). This apparent discrepancy may be explained by its attenuating effects at 300 μg/kg per se on the sympathetic vasopressor responses (Fig. 2k); this, in turn, may have overshadowed any blockade on quinpirole-induced sympatho-inhibition. In any case, within the bounds of dopaminergic mechanisms and given the selective affinity of L-745,870 for the D_4 subtype (pK_i 8.8; see Table 1), the role of this subtype in our study seems to be rather minor but cannot be categorically excluded.

Possible locus of the $D_{3/4}$ receptor subtypes mediating inhibition of the vasopressor sympathetic outflow

Although central mechanisms are not operative in our experimental model since pithed rats were used, we cannot exclude an inhibitory action of quinpirole at both the sympathetic ganglia and sympathetic perivascular nerves, which have modulatory D_2-like receptors (Wilffert et al. 1984; Willems et al. 1985; Satoh et al. 1989; Mercuro et al. 1990a, b). Admittedly, further studies in rats are required to ascertain the specific functional role of the D_2, D_3 and D_4 receptor subtypes on autonomic ganglia, but some immunohistochemical studies indicate that systemic arteries express, in a differential manner, the D_2 (pial and renal arteries), D_3 (renal artery) and D_4 (pial and mesenteric arteries) receptor subtypes displaying a prejunctional localization on sympathetic nerves (Amenta et al. 2000). In this respect, our study provides pharmacological evidence for a role of the D_3 (major)/D_4 (minor) subtypes and practically no role of the D_2 subtype, but we are measuring sympathetic vasopressor responses (reflecting systemic peripheral vasoconstriction); in contrast, Amenta et al. (2000) performed studies in different sized pial, renal and mesenteric artery branches using immunohistochemical techniques and D_1–D_5 receptor protein antibodies.

Additional findings in support of the $D_{3/4}$ receptor subtypes mediating inhibition of the vasopressor sympathetic outflow

Our findings are, indeed, consistent with other studies carried out in knockout mice. Unlike $D_2^{-/-}$ mice (whose blood pressure values are normal), $D_3^{-/-}$ mice exhibit elevated diastolic and systolic blood pressure (Zeng et al. 2008); this finding may be due, amongst others, to the D_3 subtype contribution to attenuate the sympathetic vascular tone. In this respect, it is noteworthy that nafadotride and SB-277011-A failed to potentiate the electrically-induced vasopressor responses (Fig. 2e–h), probably because the rats were systematically pretreated with desipramine before each S–R curve. In any case, this finding (a) implies that activation of D_3 receptors does not play an important role under physiological conditions, as happens with other substances such as serotonin (Villalón et al. 1995b, 1998), and (b) does not exclude the role of D_3 receptors under other circumstances, such as strenuous exercise. Indeed, several studies in humans have shown that intense exercise during acute hypoxia (Lundby et al. 2001) or without hypoxia (Mannelli et al. 1999) causes activation of sympatho-inhibitory D_2-like receptors (which include the D_2, D_3 and D_4 subtypes) that results in a decrease in circulating levels of noradrenaline.

Admittedly, other studies using $D_3^{-/-}$ mice have shown that systolic blood pressure was not different from that of wild-type animals, implying that D_3 receptors are not significantly involved in the regulation of blood pressure associated with a deficiency in renal sodium elimination (Staudacher et al. 2007). In line with these findings, administration of D_3 receptor antagonists such as A-437203 and BSF-135170 failed to increase blood pressure in conscious spontaneously hypertensive rats (Gross et al. 2006) and anaesthetised Sprague–Dawley rats (Luippold et al. 2005), respectively.

Interestingly, $D_4^{-/-}$ mice also display higher blood pressure values; however, unlike heterozygous $D_3^{+/-}$ mice, $D_4^{+/-}$ mice do not show high blood pressure values (Wang et al. 2008). This finding agrees with the minor sympatho-inhibitory role of the D_4 subtype observed in our study.

Furthermore, the physiological role of the D_3 receptor subtype is well established in the kidneys. For example, activation of the D_3 subtype has been shown to provoke (a) natriuresis and blood pressure regulation acting directly on the proximal tubule (Luippold et al. 1998) and (b) inhibition of noradrenaline release in renal nerves (Mühlbauer et al. 2000). Moreover, disruption of the D_3 receptor gene has been shown to produce renin-dependent hypertension (Asico et al. 1998). Clearly, these findings warrant further investigation to ascertain the sympatho-inhibitory role of the D_3 and D_4 receptor subtypes in cardiovascular diseases such as hypertension. In any case, our results suggest a predominant role of the D_3 receptor subtype in the inhibition of the vasopressor sympathetic outflow; this may represent another dopaminergic mechanism of blood pressure control outside the kidney.

In conclusion, the inhibition of the vasopressor sympathetic outflow induced by 1 μg/kg.min quinpirole in pithed rats is predominantly mediated by dopamine D_3 and, to a lesser extent (if any), by D_4 receptor subtypes, with no pharmacological evidence for the involvement of the D_2 subtype. This provides new evidence for understanding the modulation of the vasopressor sympathetic outflow.

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Conflict of interest The authors declare that they have no conflict of interest.

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